

Musterlösung Serie 11

LINEAR MAPS AND MATRICES

1. Matrix multiplication. Consider the real vector space $M_{2 \times 2}(\mathbb{R})$.

(a) Compute the square of

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(b) Find a formula for the entries of the n -th power of $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{N}$.

Solution:

(a) We have

$$A^2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & b(a + d) \\ c(a + d) & d^2 + bc \end{pmatrix}.$$

(b) We show by induction that for all $n \geq 1$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

For $n = 1$, we do not have anything to prove. Now let $n \geq 1$ and assume that we have proved it for all $k \leq n$. We have

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{n+1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n \cdot \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & n+1 \\ 0 & 1 \end{pmatrix},$$

where we used the induction hypothesis to obtain the second-to-last equality.

2. Linear maps as matrices. Let $\mathbb{R}[X]_n$ be the vectorspace of all polynomials of degree $\leq n$ with real coefficients.

1. Show that

$$F : \mathbb{R}[x]_n \rightarrow \mathbb{R}[x]_n, \quad p \mapsto p'' + p'$$

is a linear map, where p' denotes the derivative of p .

2. Determine the matrix of F with respect to the basis $(1, x, \dots, x^n)$ of $\mathbb{R}[X]_n$.

Solution:

1. For any $p, q \in \mathbb{R}[X]_n$, for any $\lambda \in \mathbb{R}$, we have

$$F(p + \lambda q) = (p + \lambda q)'' + (p + \lambda q)' = p'' + \lambda q'' + p' + \lambda q'.$$

Rearranging, we obtain $F(p + \lambda q) = F(p) + \lambda F(q)$. We conclude that F is linear.

2. We compute

$$F(x^i) = \begin{cases} 0, & \text{if } i = 0 \\ 1, & \text{if } i = 1 \\ i(i-1)x^{i-2} + ix^{i-1}, & \text{if } i \geq 2 \end{cases}$$

Denoting $\mathcal{B} = \{1, x, x^2, \dots, x^n\}$ the standard basis of $\mathbb{R}[x]_n$, we write it as

$$\mathcal{B}[F]_{\mathcal{B}} = ((j-1)(j-2)\delta_{i,j-2} + (j-1)\delta_{i,j-1})_{1 \leq i, j \leq n+1}.$$

In the above notation everything is offset by -1 since the power of the polynomials in the basis start with 0 but the indexing of the matrix starts with 1. More explicitly, it will look as follows:

$$\mathcal{B}[F]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 2 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 6 & \cdots & 0 \\ 0 & 0 & 0 & 3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & n(n-1) \\ 0 & 0 & 0 & 0 & \cdots & n \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

3. Matrices. Let K be a field.

- (a) Consider the matrices

$$A = \begin{pmatrix} A_1 & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & A_2 \end{pmatrix} \in M_{n \times n}(K)$$

with $A_1 \in M_{k \times k}(K)$ and $A_2 \in M_{(n-k) \times (n-k)}(K)$ for some $k \geq 1$, and

$$B = \begin{pmatrix} B_1 & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & B_2 \end{pmatrix} \in M_{n \times n}(K)$$

with $B_1 \in M_{k \times k}(K)$ and $B_2 \in M_{(n-k) \times (n-k)}(K)$. Show that

$$A \cdot B = \begin{pmatrix} A_1 \cdot B_1 & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & A_2 \cdot B_2 \end{pmatrix}$$

- (b) Let A be defined as above and assume that A_1 , respectively A_2 , is invertible as an element of $M_{k \times k}(K)$, respectively $M_{(n-k) \times (n-k)}(K)$. Show that A is invertible.
- (c) Consider the space U of upper triangular matrices in $M_{n \times n}(K)$. Show that the product of 2 elements of U is in U .

Solution:

- (a) Let us denote $C := A \cdot B$. We will write $A = (a_{ij})$, $B = (b_{ij})$ and $C = (c_{ij})$. We recall the product formula:

$$\forall 1 \leq i, j \leq n : \quad c_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.$$

Let us divide the matrices into the quadrants

$$\begin{aligned} 1 \leq i, j \leq k, & \quad 1 \leq i \leq k \wedge k < j \leq n, \\ k < i \leq n \wedge 1 \leq j \leq k, & \quad k < i, j \leq n. \end{aligned}$$

We treat the cases separately.

If $1 \leq i, j \leq k$, we have

$$c_{ij} = \sum_{\ell=1}^k a_{i\ell} b_{\ell j} + \sum_{\ell=k+1}^n 0 \cdot 0 = \sum_{\ell=1}^k a_{i\ell} b_{\ell j}.$$

The last sum is the formula to compute an entry of the product $A_1 \cdot B_1$ since i, j are arbitrary in $\{1, \dots, k\}$.

We proceed similarly for the 3 other quadrants and obtain the formula.

- (b) Denote A_i^{-1} the inverse of A_i , $i = 1, 2$ and define

$$B := \begin{pmatrix} A_1^{-1} & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & A_2^{-1} \end{pmatrix}.$$

By a),

$$A \cdot B = \begin{pmatrix} A_1 \cdot A_1^{-1} & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & A_2 \cdot A_2^{-1} \end{pmatrix} = \begin{pmatrix} I_k & 0_{k \times (n-k)} \\ 0_{(n-k) \times k} & I_{n-k} \end{pmatrix} = I_n,$$

where I_ℓ , $1 \leq \ell \leq n$, denotes the identity matrix of size $\ell \times \ell$. Hence A is invertible with inverse B in $M_{n \times n}(K)$.

- (c) Let $A, B \in M_{n \times n}(K)$ be upper triangular matrices and denote their product C . Let $1 \leq i, j \leq n$, we have

$$c_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.$$

Since $a_{i\ell} = 0$ for $\ell < i$ and $b_{\ell j} = 0$ for $\ell > j$, we have

$$c_{ij} = \sum_{\substack{1 \leq \ell \leq n \\ i \leq \ell \leq j}} a_{i\ell} b_{\ell j}, \quad \text{if } i \leq j$$

and $c_{ij} = 0$ otherwise. Hence C is upper triangular.

4. Hom as a vector space. Let V and W be finite-dimensional vector spaces over a field K .

- (a) Suppose that $2 \leq \dim V \leq \dim W$. Show that

$$\mathcal{S} := \{T \in \text{Hom}(V, W) \mid T \text{ is not injective}\}$$

is not a subspace of $\text{Hom}(V, W)$.

- (b) Suppose that $\dim V \geq \dim W \geq 2$. Show that

$$\mathcal{T} := \{T \in \text{Hom}(V, W) \mid T \text{ is not surjective}\}$$

is not a subspace of $\text{Hom}(V, W)$.

Solution:

- (a) We will show that there exist two elements T and T' of $\mathcal{S}$ such that $T + T' \notin \mathcal{S}$.

Let $\{v_1, v_2, \dots, v_r\} \subset V$ be a basis of V and let $\{w_1, w_2, \dots, w_s\} \subset W$ be a basis of W . By assumption, we have $2 \leq r \leq s$. We consider an element T of $\mathcal{S}$ defined as follows: we set $T(v_1) = w_1$ and $T(v_i) = 0$ for $2 \leq i \leq r$ and extend it linearly to obtain a linear map on the whole of V . Then $\dim(\text{Ker}(T)) = r - 1$ and $\text{rank}(T) = 1$. So, T isn't injective and T isn't the 0 map, i.e. $T \in \mathcal{S} \setminus \{0\}$.

We now define T' . We want to build it so that

$$\forall v \in V : 0 = (T + T')(v) = T(v) + T'(v) \Leftrightarrow v = 0.$$

We set $T'(v_1) = 0$, $T'(v_i) = w_i$ for all $2 \leq i \leq r$ and extend it linearly to obtain an element of $\text{Hom}(V, W)$. Assume now that $v = \sum_{i=1}^r a_i v_i \in \text{Ker}(T + T')$. Then

$$0 = (T + T') \left(\sum_{i=1}^r a_i v_i \right) = \sum_{i=1}^r a_i (T(v_i) + T'(v_i)) = \sum_{i=1}^r a_i w_i.$$

Since $\{w_1, w_2, \dots, w_r\}$ is a linearly independent subset of W , this implies $\forall 1 \leq i \leq r : a_i = 0$ and therefore, $v = 0$. This shows that $T + T' \notin \mathcal{S}$.

- (b) Let $\{v_1, v_2, \dots, v_r\}$ be a basis of V and let $\{w_1, w_2, \dots, w_s\}$ be a basis of W . This time, we have $2 \leq s \leq r$. We define $T \in \mathcal{T}$ exactly as above. It is not surjective since $\text{Im}(T) \subseteq \text{Sp}(w_1) \not\ni w_2$. We define T' on the basis as follows: we let $T'(v_1) = 0$; for $2 \leq i \leq s$, we let $T'(v_i) = w_s$; for $s < i \leq r$ we let $T'(v_i) = w_s$. We extend T' linearly to an element of $\text{Hom}(V, W)$. Since $\text{Im}(T') \subseteq \text{Sp}(w_2, \dots, w_s) \not\ni w_1$, $T' \in \mathcal{T}$. We check that $T + T'$ is surjective. Let $w = \sum_{i=1}^s a_i w_i \in W$. Then

$$\begin{aligned} w &= a_1 T(v_1) + \sum_{i=2}^s a_i T'(v_i) = \sum_{i=1}^s a_i (T(v_i) + T'(v_i)) \\ &= \sum_{i=1}^s (T + T')(a_i v_i) \in \text{Im}(T + T'). \end{aligned}$$

Hence $T + T' \notin \mathcal{T}$.

5. Change of bases. Consider the linear maps $\mathbb{R}^4 \xrightarrow{f} \mathbb{R}^2 \xrightarrow{g} \mathbb{R}^3$ given by

$$f : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + 2x_2 + x_3 \\ x_1 - x_4 \end{pmatrix} \quad \text{and} \quad g : \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + x_2 \\ x_1 - x_2 \\ 3x_1 \end{pmatrix}.$$

Let

$$\mathcal{A} := \left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 3 \\ 0 \end{pmatrix} \right),$$

let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$, and let

$$\mathcal{C} := \left(\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right).$$

1. Show that $\mathcal{A}$ is a basis of $\mathbb{R}^4$ and that $\mathcal{C}$ is a basis of $\mathbb{R}^3$.
2. Determine $g \circ f$ and the matrices
 - (i) of f with respect to the bases $\mathcal{A}, \mathcal{B}$.
 - (ii) of g with respect to the bases $\mathcal{B}, \mathcal{C}$.
 - (iii) of $g \circ f$ with respect to the bases $\mathcal{A}, \mathcal{C}$.

Lösung:

1. Gaussian elimination yields that the matrices

$$A := \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 4 & 1 & 0 \\ 1 & 2 & 1 & 3 \\ 0 & 2 & 1 & 0 \end{pmatrix} \quad \text{and} \quad C := \begin{pmatrix} 1 & 2 & 1 \\ 3 & 0 & 1 \\ 4 & 1 & 2 \end{pmatrix}$$

are invertible. This proves that $\mathcal{A}$ is a basis of $\mathbb{R}^4$ and $\mathcal{C}$ is a basis of $\mathbb{R}^3$.

2. The maps f and g can be represented by left multiplication with matrices, i.e. $f = T_U$ and $g = T_V$ for

$$U := \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix} \quad \text{and} \quad V := \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 3 & 0 \end{pmatrix}.$$

Thus we have $g \circ f = T_V \circ T_U = T_{VU}$ for the matrix

$$VU = \begin{pmatrix} 2 & 2 & 1 & -1 \\ 0 & 2 & 1 & 1 \\ 3 & 6 & 3 & 0 \end{pmatrix}.$$

The matrices U, V, VU represent $f, g, g \circ f$ with respect to the standard bases $\mathcal{B}_n$ of $\mathbb{R}^n$, which means that

$$U = [f]_{\mathcal{B}_4}^{\mathcal{B}_2}, \quad V = [g]_{\mathcal{B}_2}^{\mathcal{B}_3} \quad \text{and} \quad VU = [g \circ f]_{\mathcal{B}_4}^{\mathcal{B}_3}.$$

The matrix A is the base change matrix $A = {}_{\mathcal{B}_4}[\text{id}_{\mathbb{R}^4}]_{\mathcal{A}}$ and C is the base change matrix $C = {}_{\mathcal{B}_3}[\text{id}_{\mathbb{R}^3}]_{\mathcal{C}}$.

- (i) The matrix representing f w.r.t. $\mathcal{A}$ and $\mathcal{B}$ is

$$\begin{aligned} [f]_{\mathcal{A}}^{\mathcal{B}} &= [f]_{\mathcal{B}_4}^{\mathcal{B}} \cdot [\text{id}_{\mathbb{R}^4}]_{\mathcal{A}}^{\mathcal{B}_4} \\ &= U \cdot A \\ &= \begin{pmatrix} 2 & 11 & 4 & 5 \\ 1 & -1 & 0 & 2 \end{pmatrix}. \end{aligned}$$

- (ii) We have

$$\begin{aligned} [g]_{\mathcal{B}}^{\mathcal{C}} &= [\text{id}_{\mathbb{R}^3}]_{\mathcal{B}_3}^{\mathcal{C}} \cdot [g]_{\mathcal{B}}^{\mathcal{B}_3} \\ &= ([\text{id}_{\mathbb{R}^3}]_{\mathcal{C}}^{\mathcal{B}_3})^{-1} \cdot [g]_{\mathcal{B}}^{\mathcal{B}_3} \\ &= \begin{pmatrix} -1 & -1 \\ -1 & 0 \\ 4 & 2 \end{pmatrix}. \end{aligned}$$

(iii) The matrix representing $g \circ f$ w.r.t $\mathcal{A}$ and $\mathcal{C}$ is

$$\begin{aligned} [g \circ f]_{\mathcal{A}}^{\mathcal{C}} &= [g]_{\mathcal{B}}^{\mathcal{C}} \cdot [f]_{\mathcal{A}}^{\mathcal{B}} \\ &= \begin{pmatrix} -3 & -10 & -4 & -7 \\ -2 & -11 & -4 & -5 \\ 10 & 42 & 16 & 24 \end{pmatrix}. \end{aligned}$$

6. Linear maps. Let V be a vector space over a field K . Suppose that T_1 and T_2 are two linear maps from V to K that have the same kernel. Show that there exists a constant $c \in K$ such that $T_1 = cT_2$.

Hint: Use Serie 10 exercise 1.

Solution: If T_1 or T_2 is the 0-map, we are done. Assume that T_2 isn't the 0-map, i.e. that $\ker(T_2) \subsetneq V$. Let $u \in V \setminus \ker(T_2)$. Then, $v_0 := \frac{u}{T_2(u)}$ is such that $T_2(v_0) = 1 \in K$. Let $v \in V$ and consider

$$w = v - T_2(v)v_0 \in V.$$

It is an element of $\ker(T_2)$ since

$$T_2(w) = T_2(v) - T_2(v)T_2(v_0) = T_2(v) - T_2(v) = 0.$$

We assumed that $\ker(T_1) = \ker(T_2)$, therefore

$$0 = T_1(w) = T_1(v) - T_2(v)T_1(v_0) \Leftrightarrow T_1(v) = T_1(v_0)T_2(v).$$

Since $v \in V$ was arbitrary, we set $c := T_1(v_0)$ and conclude that

$$T_1 = cT_2.$$

7. Linear maps as matrices. Denote $\mathbb{R}[x]_d$ the set of polynomials over $\mathbb{R}$ of degree lower or equal to d . Suppose that $D \in \text{Hom}_{\mathbb{R}}(\mathbb{R}[x]_3, \mathbb{R}[x]_2)$ is the differentiation map $Dp = p'$. Find a basis of $\mathbb{R}[x]_3$ and a basis of $\mathbb{R}[x]_2$ such that the matrix of D with respect to these bases is

$$\begin{pmatrix} 0 & 1 & -1 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Solution: Consider the basis $\{1, 1+x, 1+x+x^2, 1+x+x^2+x^3\}$ of $\mathbb{R}[x]_3$ and the basis $\{1, 1+x, 1+x+x^2\}$ of $\mathbb{R}[x]_2$. We have

$$\begin{aligned} D(1) &= 0 &= 0 \cdot 1 + 0 \cdot (1+x) + 0 \cdot (1+x+x^2) \\ D(1+x) &= 1 &= 1 \cdot 1 + 0 \cdot (1+x) + 0 \cdot (1+x+x^2) \\ D(1+x+x^2) &= 1+2x &= -1 \cdot 1 + 2 \cdot (1+x) + 0 \cdot (1+x+x^2) \\ D(1+x+x^2+x^3) &= 1+2x+3x^2 &= -1 \cdot 1 + -1 \cdot (1+x) + 3 \cdot (1+x+x^2) \end{aligned}$$

Hence the matrix of D is the desired one with respect to these two bases.

Multiple Choice Questions. More than one answer can be correct.

Question 1. Let $\mathcal{A}$ and $\mathcal{B}$ be bases of $\mathbb{R}^2$ and denote $\{e_1, e_2\}$ the standard basis. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map given by the matrix

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

with respect to $\mathcal{A}$ as a basis of the domain and $\mathcal{B}$ as a basis of the codomain. Which of the following statements are true?

- ✓ If $\mathcal{A} = \mathcal{B} = \{e_1, e_2\}$, f is a rotation around the origin.
- ✓ If $\mathcal{A}$ is the standard basis and $\mathcal{B} = \{e_2, -e_1\}$, f is a symmetry with respect to the point $(0, 0)$.
- ✓ If $\mathcal{A}$ is the standard basis and f is the identity, then $\mathcal{B} = \{-e_2, e_1\}$.
- ✓ If $\mathcal{B}$ is the standard basis and f is the symmetry with respect to the y -axis, then $\mathcal{A} = \{e_2, e_1\}$.

Question 2. Which of the following statements are true?

- ✓ Let V be an n -dimensional vector space over $\mathbb{R}$. The map $V \rightarrow \mathbb{R}^n$, $v \mapsto [v]_{\mathcal{B}}$, that sends any vector $v \in V$ to its coordinate vector with respect to a basis $\mathcal{B}$ of $\mathbb{R}^n$ is linear.
- Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear with $\text{Ker}(f) \neq \{0\}$ and $\text{Im}(f) \neq \{0\}$. Then there exists a non-trivial vector v that is in the kernel and in the image of f .
- ✓ If the kernel of a linear map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is trivial, the map is invertible.